AIAA SciTech Forum
January 3-7, 2022, San Diego, CA & Virtual
AIAA SCITECH 2022 Forum
10.2514/6.2022-1027

Development of a Multi-Purpose Flap and Spoiler Mechanism for High Endurance Unmanned Aerial Vehicles

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High endurance aircraft can provide benefits to many operations. However, sub-optimal flap mechanisms hinder takeoff and landings, especially on small unmanned aerial vehicles (UAVs). High-lift mechanisms could greatly improve small UAVs' takeoff and landing performance but are not implemented due to their weight and complexity. Hence, there is a need for a simple control surface mechanism that can provide aileron, spoiler, and high-lift flap functionalities. Three novel designs were proposed: The Extended Flaps and Airbrakes (EXFA) System, and two variants of the High-Lift EXFA (HEXFA) System, one with a smooth upper control surface hinge and the other with retractable vortex generators. Investigations into the performance of these systems were conducted experimentally in the University of Dayton Low Speed Wind Tunnel (UD-LSWT) and computationally through SU2. The aerodynamic coefficients in each configuration were analyzed and compared with traditional flap configurations. Manual and autonomous UAV flight tests with the EXFA System were performed to validate the integrability of these systems in a full R/C aircraft system. The results suggest that the HEXFA System exhibits comparable lift characteristics to a slotted Fowler flap at low angles of attack, while being able to effectively function as an aileron and spoiler, thus offering significant weight advantages. The flight tests corroborated the effectiveness of the EXFA System at shortening landing distance and substantiated its integrability in autonomous flight.

I. Nomenclature

α	=	angle of attack (degrees)
C_L	=	lift coefficient
C_D	=	drag coefficient
c	=	chord length (m)
τ_x	=	torque around the x axis, rolling moment (Nm)
D	=	drag force (N)
L	=	lift force (N)
S	=	wing planiform area (m^2)
U_∞	=	freestream velocity (m/s)

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ρ	=	air density (kg/m^3)
θ	=	pitch angle relative to the horizon (degrees)
g	=	acceleration due to gravity (m/s^2)
ϕ	=	complementary angle to the pitch angle (degrees)
a_x	=	measured acceleration along the longitudinal or forward and backward axis of the aircraft (m/s^2)
m	=	mass of aircraft (kg)
RPM	=	rotations per minute (1/min)
d	=	horizontal approach distance (m)
V_{plane}	=	airspeed of aircraft (m/s)
$\bar{V}_{plane}$	=	average airspeed of aircraft (m/s)
t	=	time (s)
σ	=	standard deviation

II. Introduction

High endurance aircraft can provide benefits to a variety of operations including intelligence gathering, close air support [1], and ship-based ocean research [2]. However, a major drawback of high-endurance aircraft is their long takeoff and landing distances [3]. Utilizing either catapults and recovery systems or vertical takeoff and landing (VTOL) capabilities is an alternative solution to shorten landing distance [2]. However, utilizing catapult and recovery systems constrains takeoff and landing locations while VTOL mechanisms often add weight and complexity to the aircraft, reducing its flight endurance.

Oliveira [4] investigated the design and effectiveness of a slotted fowler flap system on an airfoil optimized for low Reynolds number operation (Re = 200,000) of a small UAV. The results of the computational fluid dynamics simulations (Fig. 1) exemplify the high-lift performance of the slotted Fowler flap, which provided a C_L increment of over 1.5 for most cases when compared to the baseline [4]. However, the actuation mechanism for the slotted Fowler flap is complex, requiring multiple joints and thin load-bearing links, which would add weight to the aircraft and be hard to integrate into the small geometry of a UAV wing. The system is also only able to function as a flap and not as an aileron or drag increasing spoiler, which limits its functionality.

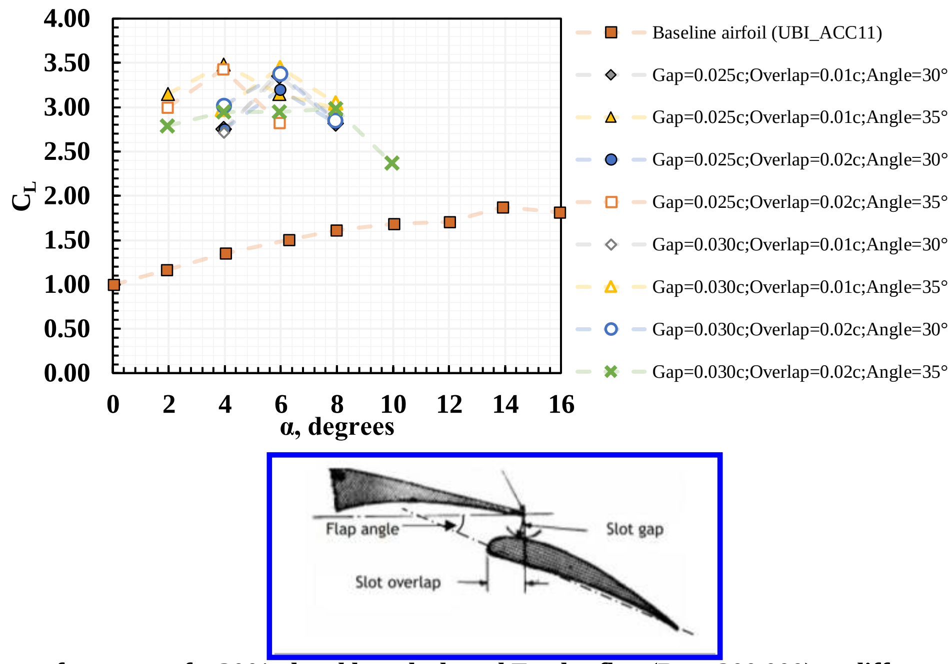

Fig. 1 C_L performance of a 30% chord length slotted Fowler flap (Re = 200,000) at different deployment angles [4]

On the other hand, active flow control with sweeping jet actuators has been shown to enhance the lift generation of a hinged flap. Melton et al. [5] experimentally investigated the influence of sweeping jet (SWJ) activators and convergent-divergent (CD) actuators on a hinged flap at low Reynolds number (Re = 200,000-700,000). Their results (Fig. 2a) demonstrate significant C_L increment for all flap deflections, with C_L increments of over 0.6 for the SWJ actuators at high mass flow rates [5]. Hinged flaps can also be utilized for aileron functionality but are still unable to provide drag increasing functionality. Active flow control mechanisms also add significant structural complexity and require compressors and other devices to achieve effective momentum flow coefficients, increasing weight and reducing integrability in small UAVs.

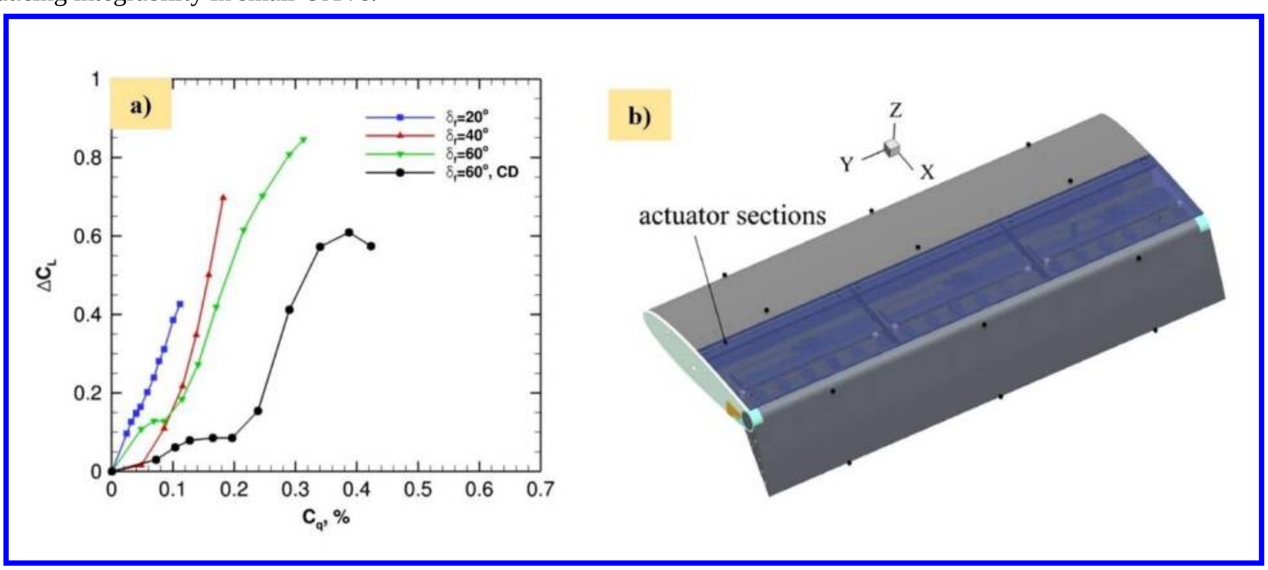

Fig. 2 a) Lift increment as a function of mass flow coefficient (C_q) for different flap angles (δ_f) and b) wing and actuator system tested [5]

Other methods for increasing lift on takeoff have been explored, including trailing edge flap geometries, leading edge slats and extensions [6], wing morphing [3], and vortex generators [7]. However, these systems either are mechanically complex, which increases weight, or only have marginal benefits. Hence, many high-endurance aircraft house sub-optimal high lift devices. Without high lift devices, the aircraft are forced to takeoff and land at higher speeds, requiring longer runways and landing approach paths [3].

The most effective methods of reducing landing distance are the use of wheel brakes and flaps [6]. Despite their effectiveness on the ground, brakes have no contribution in the air. Flaps can reduce the stall speed of the aircraft or add drag in air, allowing it to descend at a steeper angle and touch down at a slower speed [6]. Two other mechanisms designed to shorten landing distance are spoilers and airbrakes, both by increasing the sectional drag of the wing.

Despite the benefit and utility of augmenting ailerons with spoilers and high-lift flap systems, these are usually integrated into aircraft as separate systems. Although some aircraft combine two of the functions into a single system, due to the mechanical complexity and subsequently added weight of adding any of these systems, these mechanisms are often not implemented on high endurance aircraft [3]. Thus, there is a need for a simple and innovative control surface mechanism that combines the function of high-lift device, aircraft control and speed break, which would allow the single mechanism to cover a larger section of the wing, leading to a higher effectiveness of each system function. In addition, minimizing the design complexity subsequently could reduce the weight of the system, minimizing the performance reduction of the aircraft.

III. Test Models and Experimental Setup

A. EXFA and HEXFA System Design Concepts

Three novel control surface mechanisms were proposed, which could combine flap, spoiler, and airbrake functionalities into a two-control-surface mechanism. The first design was the Extended Flaps and Airbrakes (EXFA) System, shown in Fig. 3a. The EXFA System consists of an upper control surface and a lower control surface that can be independently actuated to both upward and downward control surface deflection angles. The EXFA System can be deployed to four positions. Besides the neutral position, the EXFA System can be actuated to the flap position, in which both upper and lower control surfaces are actuated downwards, resembling a hinged flap. The EXFA System can also be deployed to the aileron position, with both control surfaces angled upwards. Both functions resemble a flaperon system. Additionally, the EXFA System can be deployed to the spoiler position in which the upper control surface is deflected upwards, while the lower control surface is deflected downwards. The net result of this deployment is a decrease in lift and an increase in drag, similar to the function of an airbrake.

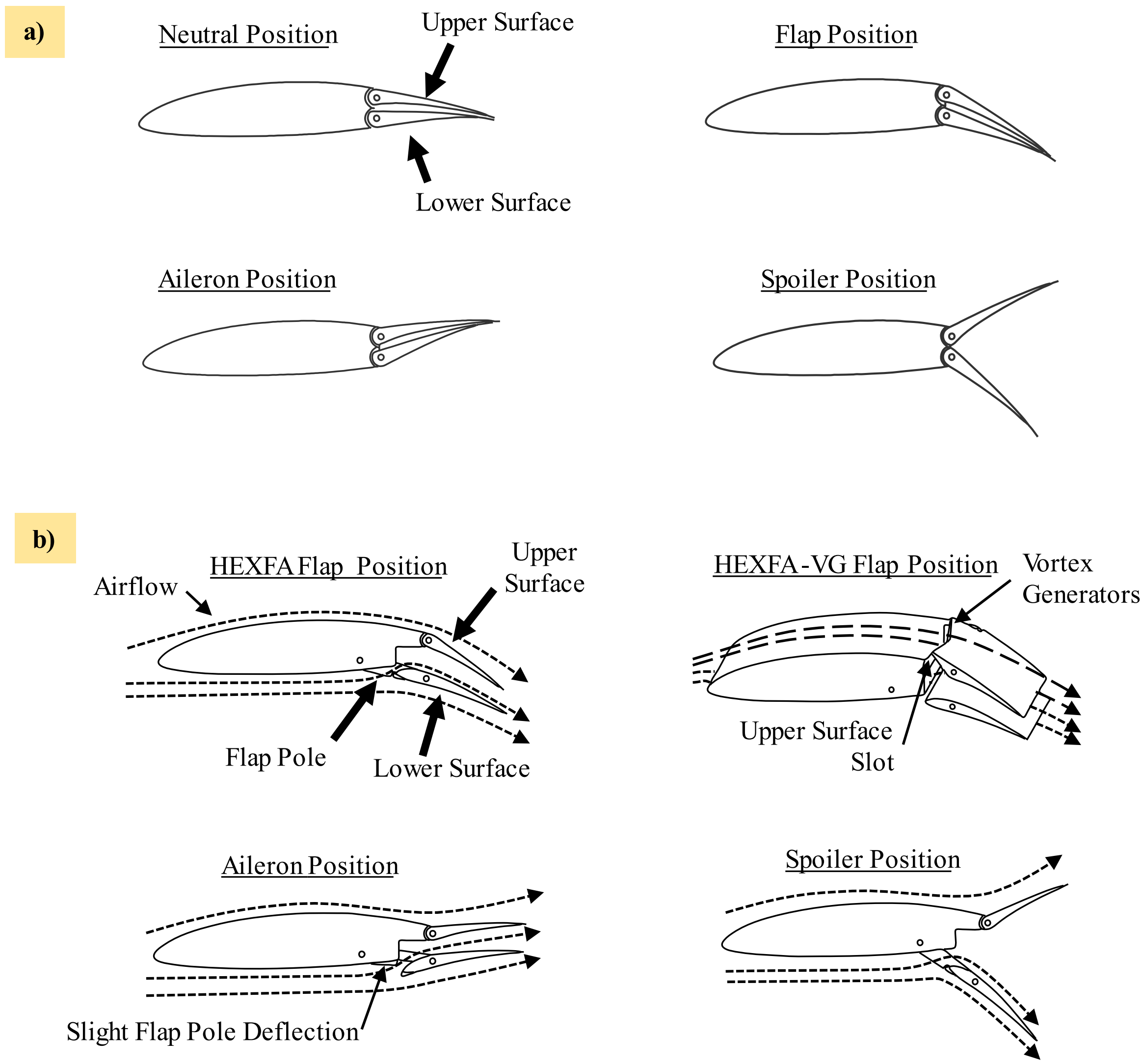

Fig. 3 Design of a) EXFA System and b) HEXFA and HEXFA-VG Systems.

The design of the High Lift Extended Flap and Airbrake (HEXFA) System, shown in Fig. 3b, focused on increasing the lift of the EXFA System in the flap position, while retaining the aileron and spoiler functions. For the lower control surface, a slot was added to allow air to flow over the top of itself, similar to a slotted flap, which could increase lift, especially at high angles of attack [6].

To attach the slotted flap to the wing, flap poles hinged at both ends were utilized. The main hinge connects the main wing to the flap poles, and the flap hinge connects the flap poles to the lower control surface. When the HEXFA System is deployed to the flap position, the flap control surface and flap poles rotate around the main flap hinge, exposing the slot.

Two versions of the HEXFA System were designed for increased the lift on the main section of the wing. The first version, referred to as simply the HEXFA System, maintained a smooth and straight junction between the main upper surface of the wing and the upper control surface. The second version is the HEXFA System with vortex generators (HEXFA-VG System), which utilizes retractable vortex generators on the upper control surface. In the flap position, the upper control surface is deflected downwards, exposing vortex generators on the leading edge, aiming to increase the flap system performance.

For the aileron position, both the HEXFA System and the HEXFA-VG System operate identically. The upper control surface is deflected upwards like a normal aileron and the lower control surface is also rotated upwards around the flap hinge, exposing the leading edge into the freestream The HEXFA and HEXFA-VG Systems operate identically in the spoiler position as well. The lower control surface is rotated downwards around the main hinge, and the upper control surface is deflected upwards.

B. Wind Tunnel Testing Models

Six wing models were fabricated for the study: a baseline wing, hinged flap wing, slotted Fowler flap wing, EXFA wing, HEXFA wing, and HEXFA-VG wing (Fig. 4). All wings had a semi-span aspect ratio of 2.5 with a chord length of 12 cm and a wingspan of 30 cm. A NASA NLF 1015 airfoil geometry was used for all the wing models. The wings were designed using Dassault Systems SolidWorks 2020-2021 Student Edition (SolidWorks) and 3D printed on an Ender 5 fused filament fabrication (FFF) 3D printer with polylactic acid (PLA) filament.

No control surfaces were added for the baseline wing (Fig. 4a). 0.82mm piano wire was used on all of the hinges on the hinged flap, EXFA, HEXFA, and HEXFA-VG wings, and a flap track system was used on the slotted Fowler flap wing (Fig. 5). All flap system wings had 33% wing chord length control surfaces with a maximum downward flap deflection of 40°.

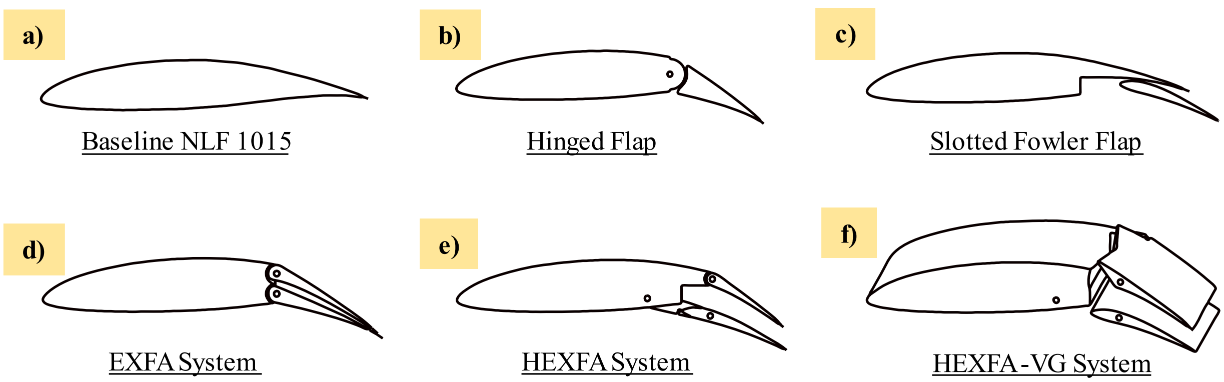

Fig. 4 Cross sections of baseline and flap systems for wind tunnel testing.

The control surfaces on the hinged flap, slotted Fowler flap, EXFA, HEXFA, and HEXFA-VG wings were actuated by 9-gram servos. The servos were placed in slots on the wing and covered with tape, only the control horn was exposed to the freestream. A computer graphical user interface (GUI) was designed to wirelessly control the servos through a wireless serial communication connection. An Arduino microcontroller was used to receive the signal from the computer and send the position commands to the servos.

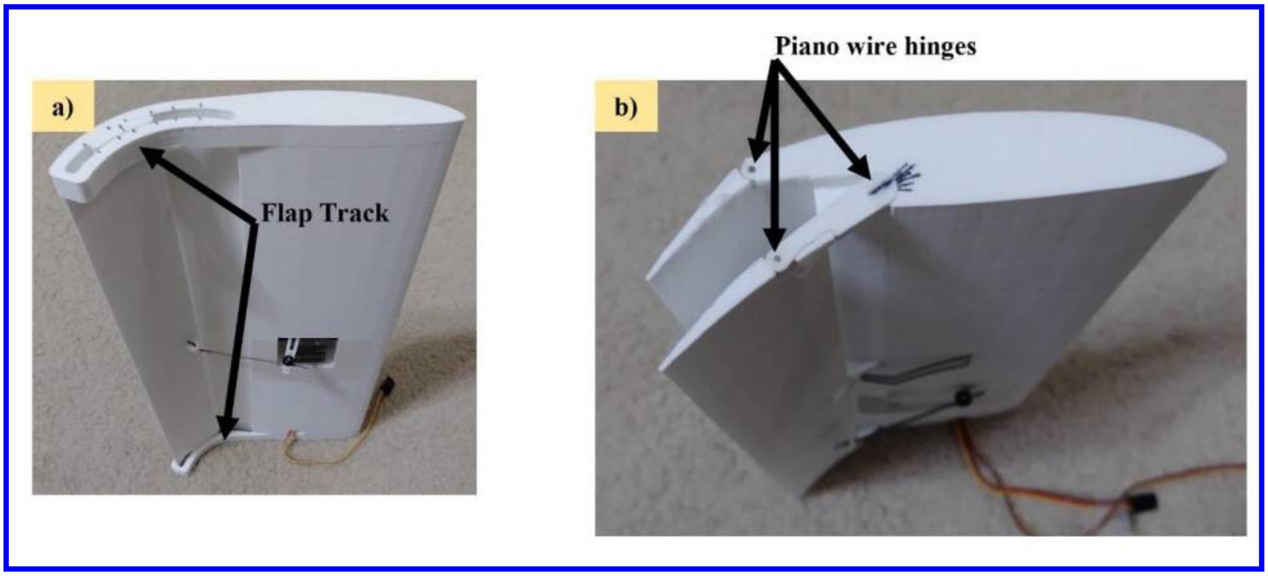

Fig. 5 Pictures of a) slotted Fowler flap and b) HEXFA System wind tunnel wings

C. Wind Tunnel Experimental Setup

All wings were tested in the University of Dayton Low-Speed Wind Tunnel (UD-LSWT) in the open jet configuration. The wind tunnel has an inlet contraction ratio of 16:1, a 762 mm x 762 mm testing section inlet and a 1370 mm x 1370 mm collector. The freestream velocity ranges from 3 m/s to 37 m/s, with a turbulence intensity of 0.1% measured by a hot wire anemometer at 15 m/s.

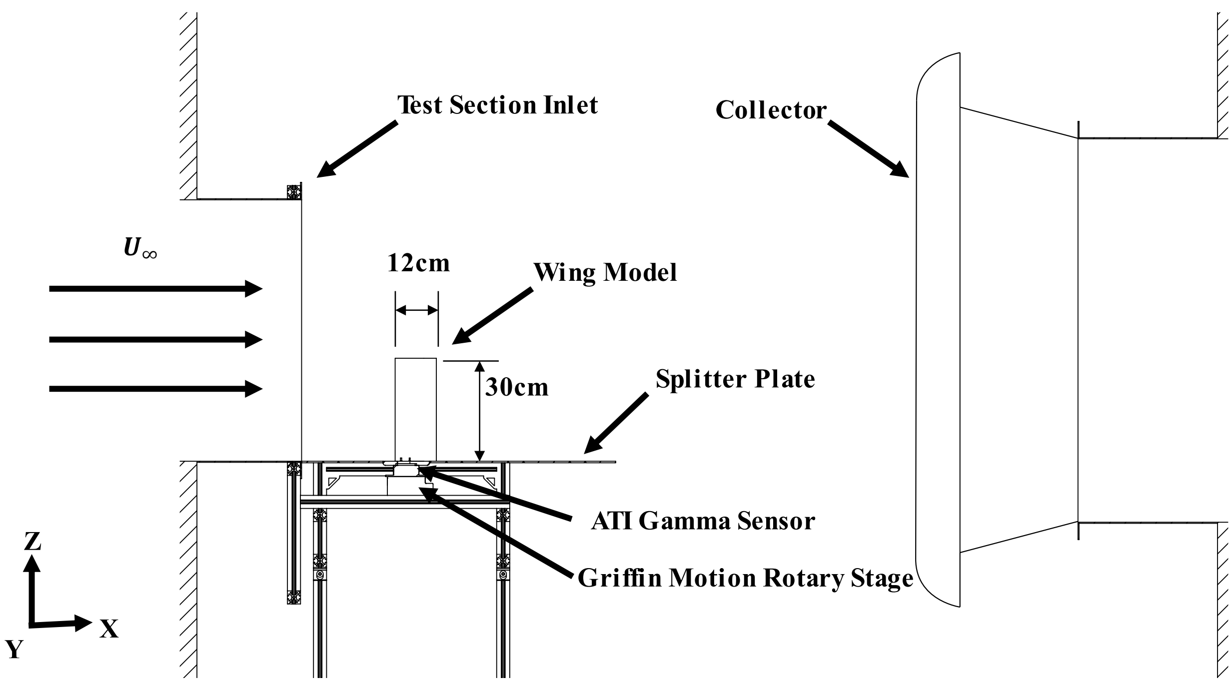

Fig. 6 University of Dayton Low-Speed Wind Tunnel (UD-LSWT) test setup.

The wing models were mounted to the ATI [8] Gamma F/T transducer with a splitter plate covering one wingtip as shown in Fig. 6. The sensor has a force resolution of 1/80 N and range of 65 N on the x and y axes and a force resolution of 1/40 N and range of 200 N on the z axis. The sensor has a torque resolution of 1/1333 Nm and range of

5 Nm on all axes. The sensor is mounted on a Griffin motion rotary stage to change the angle of attack of the sensor and the wing. LabVIEW was used for motion control and data acquisition.

Table 1 shows the testing matrix for the wind tunnel experiment. The lift force for each wing at 0° angle of attack with 0° flap deflection was matched to the baseline lift force at 0° angle of attack. After matching the lift, the control surface is deflected, for the performance measurement. Due to the torque limitation of the servo, the maximum freestream velocity achieved for the flap position is 20 m/s (Re = 170,000), while the maximum freestream velocity for aileron and spoiler position is 15 m/s (Re = 127,000). The wings were carefully observed and the data was retaken if there was any adverse control surface deflection. Only practical angle of attack range is tested for each configuration. 500 samples of each force and moment component were taken over a period of around 5 seconds. The mean and standard deviation were calculated for each value.

Table 1 Wind tunnel testing matrix

	α Range	α increment	$\boldsymbol{U_\infty}$	Wing Models Tested	Deployment Angle
Flap Position	-2°–16°	1°	20 m/sec	Baseline (NASA NLF-1015), Plain Flap, Slotted Fowler, EXFA, HEXFA, HEXFA-VG	0°, 20°, 40° Deflection
Aileron Position	0°–10°		15 m/sec	Plain Flap, EXFA, HEXFA, HEXFA-VG	0°, 10°, 20° Deflection
Spoiler Position	0°–10°		15 m/sec	EXFA, HEXFA, HEXFA-VG	0°, 20°, 40° Deflection

The lift and drag force data sets were converted to lift coefficient (C_L) and drag coefficient (C_D) data using Eqs. (1) and (2):

$$C_L = \frac{L}{0.5\rho(U_\infty)^2 S} \tag{1}$$

$$C_D = \frac{D}{0.5\rho(U_\infty)^2 S} \tag{2}$$

The C_L and C_D were calculated individually for all 500 samples in each test to extract the mean and standard deviation values of the C_L and C_D. For the aileron and spoiler cases, the change in rolling moment ($\Delta\tau_x$) and change in drag coefficient (ΔC_D) were calculated for each control surface deflection angles by subtracting the moment and drag in the deployed configuration from the neutral configuration as shown in Eqs (3) and (4). The mean and standard deviation values were calculated individually for the deployed and neutral configurations. The standard deviation of $\Delta\tau_x$ and ΔC_D were calculated by combining the individual standard deviations according to [12], shown in Eq. 5:

$$\Delta\tau_x = |\tau_{xNeutral}(\alpha) - \tau_{xDeployed}(\alpha)| \tag{3}$$

$$\Delta C_D = |C_{DNeutral}(\alpha) - C_{DDeployed}(\alpha)| \tag{4}$$

$$\sigma_{combined} = \sqrt{\sigma_{Neutral}^2 + \sigma_{Deployed}^2} \tag{5}$$

D. Computational Fluid Dynamics (CFD) Simulation

Initially, the airfoil geometries were created by taking a planar cross section of the SolidWorks 3D models at a spanwise location that retained the optimal airfoil geometry for that wing. Subsequently, the airfoil section was imported into GMSH to create the simulation meshes (Fig. 7). The mesh generated was an unstructured 2D meshes with triangular mesh elements. The SU2 Multiphysics Simulation and Design Software was used to perform CFD simulations. A farfield boundary condition was utilized for the edges of the mesh (Fig. 7a) and an adiabatic, no-slip wall boundary condition was utilized for the airfoil geometry (Fig. 7b). From a grid convergence study of this simulation setup with meshes of the baseline airfoil with node counts ranging from around 170 K to 1M nodes, the mesh C_L error when compared to the finite mesh was found to be 4.0%. The testing conditions and airfoils tested are listed in Table 2. To compare the 2D CFD results with the wind tunnel data, aspect ratio correction was performed according to [6], shown in Eq.6:

$$C_L = C_l \cdot \left(\frac{AR}{AR + 2\frac{(AR+4)}{(AR+2)}} \right) \tag{6}$$

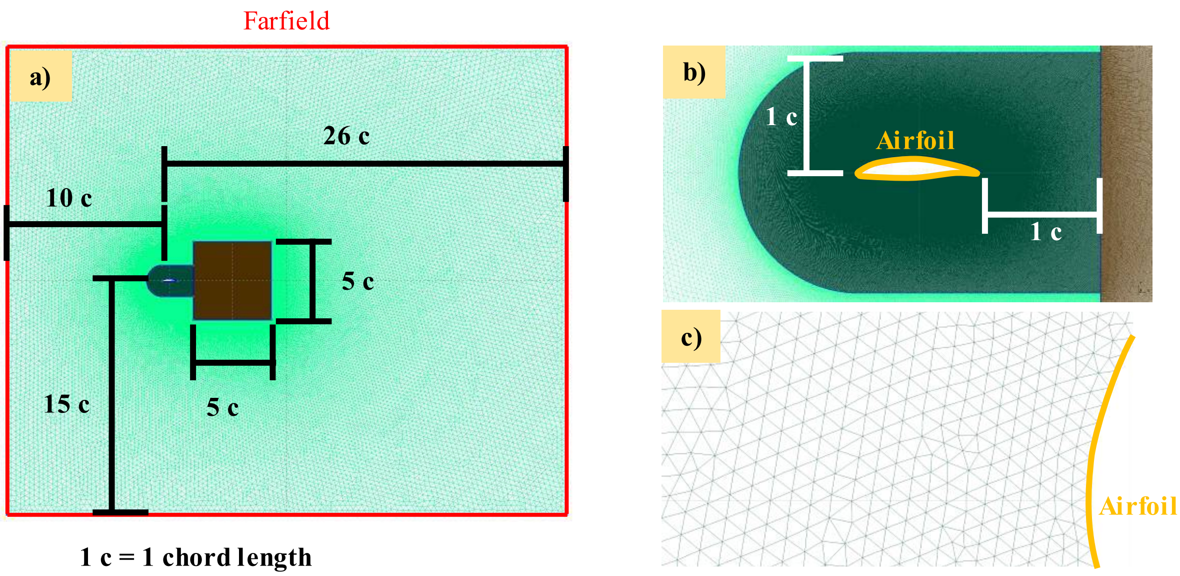

Fig. 7 View of a) full mesh, b) nearfield region, and c) close up of airfoil leading edge for a ~10^6 node baseline mesh with farfield and airfoil boundaries labeled.

Table 2 CFD testing matrix

α Range	U_∞	Software	Node Count	Turbulence Model	2D Airfoils Tested	Deployment Angle
-2°–10°	20 m/sec	SU2	1,000,000 nodes (~1 Y+)	Spalart-Allmaras	Baseline, Slotted Fowler, HEXFA	20° Flap Position Deflection

E. UAV Flight Tests
1. Test Aircraft

Two high-endurance unmanned aerial vehicles were used for the flight test for the EXFA system. The first aircraft, referred to as the "EXFA pusher aircraft", is shown in Fig. 8a and b. The UAV was constructed mostly out of foam board, carbon fiber rods, and wood, with an overall mass of 1.2 kg, wingspan of 1.8 m, and average chord length of 0.15 m (AR = 12). The EXFA control surfaces covered 66% of the chord length and 40% of the wingspan (Fig. 8c). A standard aileron was installed on the outboard section of the wings for roll control. The UAV was manually piloted from the ground with a Spectrum DX7 remote control transmitter via a 2.4Ghz radio link. The control surfaces on the aircraft were actuated by 9 gram servos and the aircraft was powered by a 200 W motor with an 8 in (20 cm) pusher propeller.

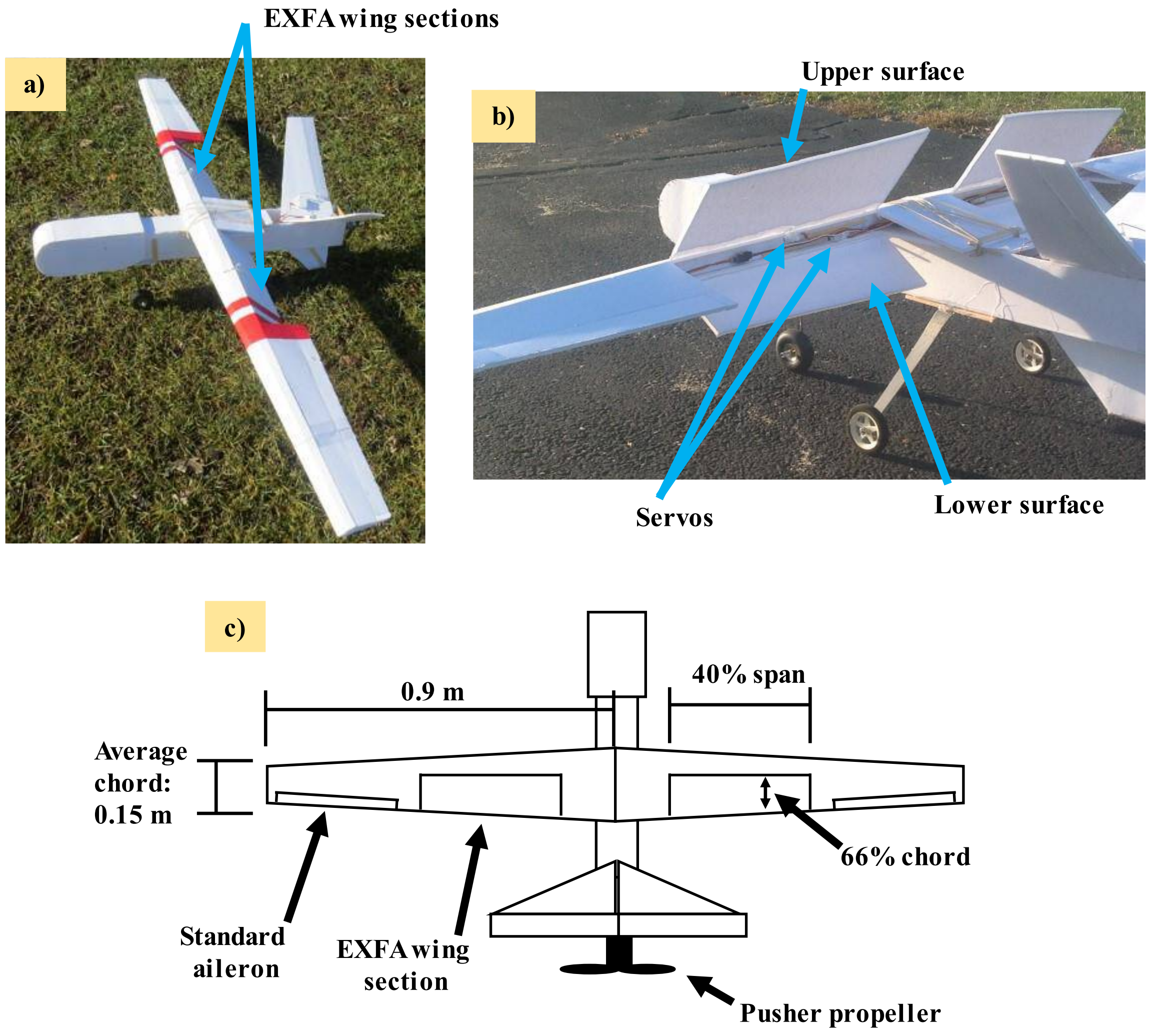

Fig. 8 a) picture of full EXFA pusher aircraft, b) picture of EXFA wing section, and c) illustration of aircraft setup

The other high-endurance UAV used for flight testing is shown in Fig. 9a and was referred to as the "EXFA tractor aircraft". The aircraft was built with an overall mass of 3.4 kg, wingspan of 2 m, and chord length of 0.22 m (AR = 9). Aside from the metal landing gear, the fuselage and tail of the aircraft were constructed primarily out of G10 epoxy and carbon fiber composites. The control surfaces on the aircraft were actuated by 18-gram servos and the aircraft was powered by a 600 W motor with a 12 in (30.5 cm) tractor propeller. The EXFA System control surfaces covered 45% of the wing chord and 33% of the wingspan (Fig. 9b). A standard aileron was installed on the outboard section of the wings for roll control (Fig. 9c). This aircraft was equipped with a Pixhawk 2.1 autonomous flight controller system. The flight controller and global positioning system (GPS) modules were mounted inside the fuselage near the aircraft's center of gravity (Fig. 9d) and the pitot tube was imbedded in the wing (Fig. 9c).

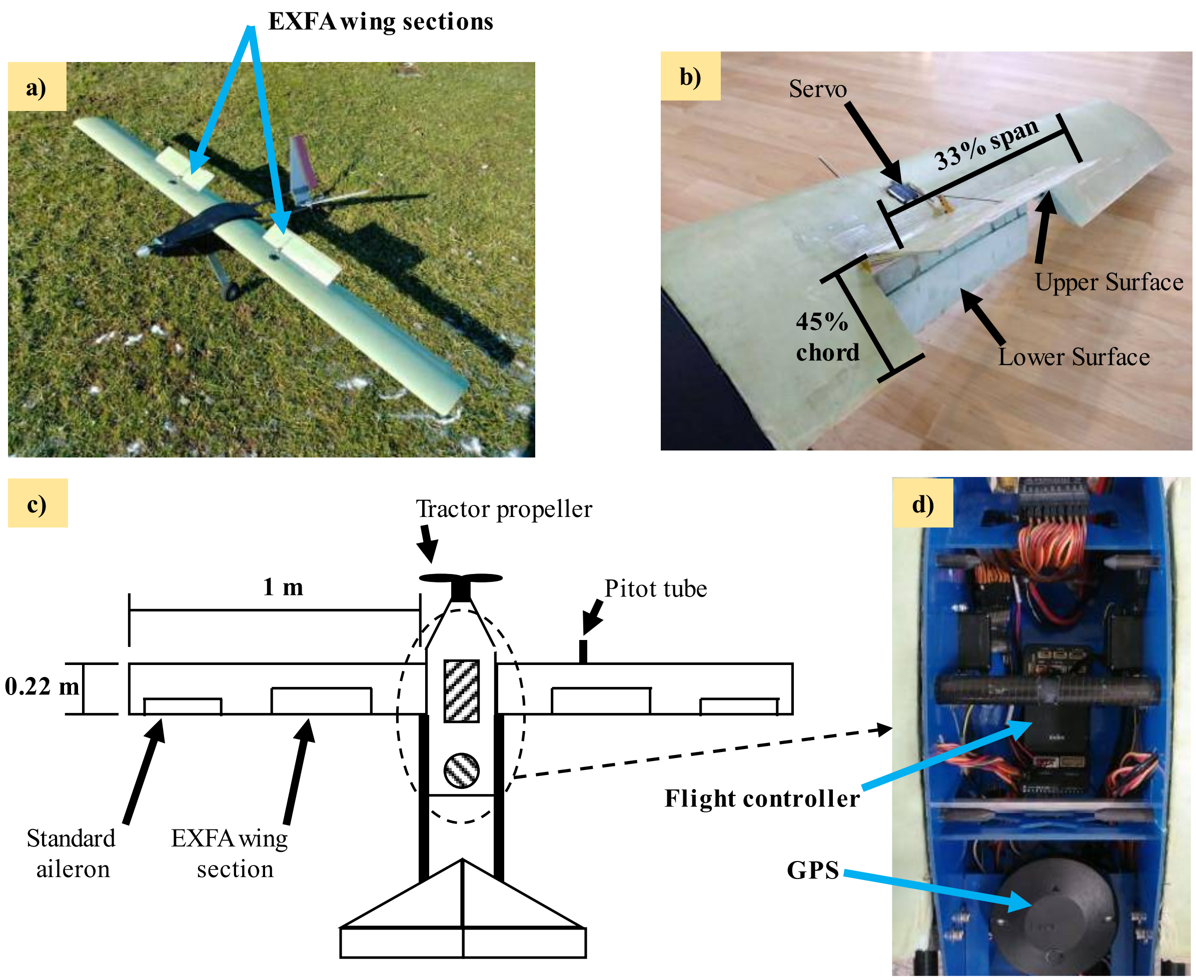

Fig. 9 a) picture of full EXFA tractor aircraft, b) picture of EXFA wing section, c) illustration of aircraft setup, and d) picture of flight controller and GPS placement in fuselage

2. Estimating Drag

The EXFA tractor aircraft was used to estimate the drag data. The Pixhawk 2.1 flight controller was utilized to measure and record the acceleration of the aircraft. Initially, the RC aircraft was flown manually up to a suitable height, trimmed for cruise condition, the pilot reduced the throttle to zero and put the aircraft into a dive, increasing the downward pitch angle of the dive gradually to sixty degrees, and then leveled the aircraft again. An illustration of a flight pass is shown in Fig. 10. This captured the forward acceleration of the aircraft for every downward pitch angle between zero and sixty degrees. This process was repeated for a total of five times for each EXFA System position (neutral, 40° flap deflection, 40° aileron position deflection, and 40° spoiler position deflection).

Fig. 10 Illustration of flight pass for drag data collection

To test the accuracy of the sensors, a different setup was utilized. The aircraft was placed in a stationary position for 108 seconds while the servos were actuated to simulate the electronic noise during the flights, and the sensor data was recorded for that test. The standard deviation (95% confidence) of the sensor data was taken to determine the accuracy of the sensors. The accuracy of the aircraft's pitch sensor was found to be ± 0.068 degrees and the accuracy of the acceleration in the x-dimension (the longitudinal or forward and backward axis) was found to be ± 0.026 m/sec^2.

A free body diagram of the forces on the aircraft and flight controller is shown in Fig. 11. Since the motor was off, the only forces acting along the x-axis of the aircraft were gravity and drag. Using the pitch angle of the aircraft and the measured acceleration (a_x), the drag force was calculated using Eq. 7. The angle of attack was calculated from the inertial measurement unit and airspeed sensor data using calculations based on [9], [10], and [11]. The C_D was also calculated for the neutral position (Eqs. 8-9) using the planar surface area (S) of the entire wingspan. This calculation was valid at 0° angle of attack (α), where the acceleration readings (a_x) were colinear with the drag force and was a close approximation for low angles of attack.

$$D = (m)\big((g)cos(\phi) - a_x\big) \quad (7)$$

$$C_D = \frac{D}{0.5\rho V^2 S} \quad When\, \alpha = 0° \quad (8)$$

$$C_D \approx \frac{D}{0.5\rho V^2 S} \quad When - 5° < \alpha < 5° \quad (9)$$

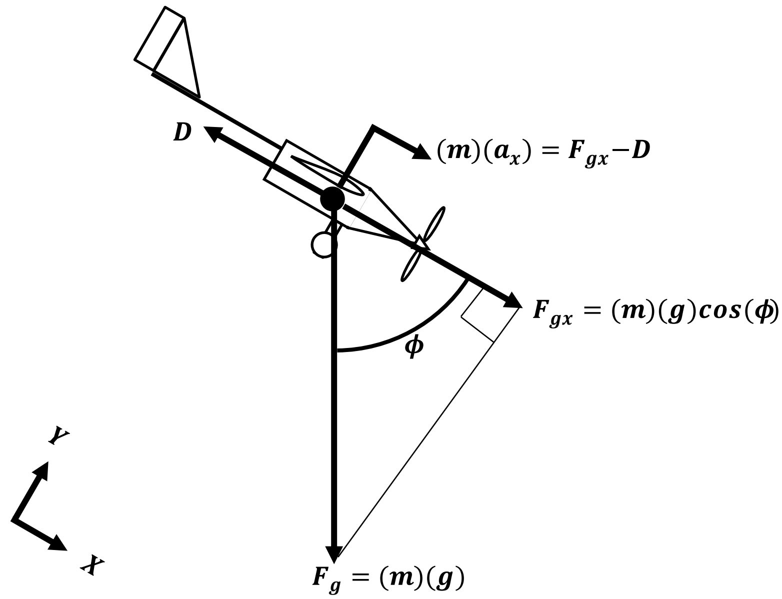

Fig. 11 Free body diagram of the aircraft

The change in C_D (ΔC_D) was then calculated for the deployed EXFA positions. The trendline of the neutral position C_D data was taken to represent the neutral position Cd as a function of angle of attack. For the deflected positions, the angle of attack and airspeed measurements from the flight controller was used to calculate the neutral position drag force under the conditions for each data point. The change in drag force from the EXFA System was calculated by taking the difference between the measured deflected drag force and the calculated neutral position drag force and converted to change in Cd using Eqs. 6 and 7, with S as the planar area of the EXFA System wing sections.

3. Estimating Landing Distance

To collect landing distance data, flight tests were conducted with the EXFA pusher aircraft as shown in Fig. 12. Two observers were positioned on the initial half of the runway. The observers watched the aircraft and marked where the main wheels touched the ground. After the aircraft came to a complete stop, the observers marked where the main wheels stopped and then measured the distance between the two markers. This method was used for collecting the landing distances. Each landing was video recorded, and the ground roll distance and wind speed were measured. Landings were conducted for three control surface deployments: 1. landings without deploying the EXFA System, 2. landings with only the EXFA flap position (~40° deflection angle), and 3. landings utilizing the whole EXFA system, with EXFA flap deployed (~40° deflection angle) in the air and EXFA spoiler deployed (~40° deflection angle) immediately after the wheels touch the ground.

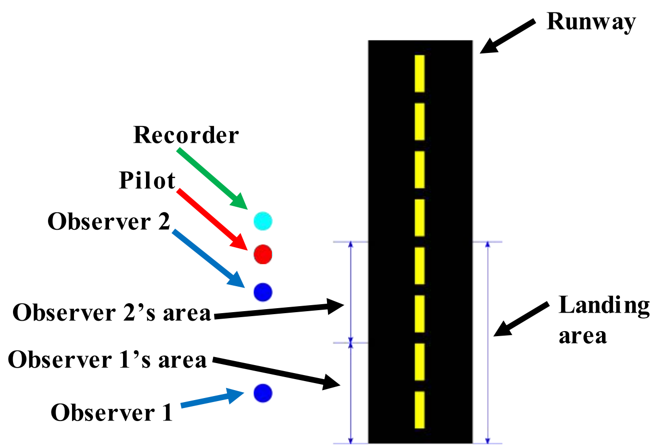

Fig. 12 schematic of setup for landing distance data collection.

Criteria for an acceptable landing was defined as a landing where: 1. the aircraft approached the runway at the slowest possible controlled landing speed for the given EXFA deflection, 2. the aircraft touched the ground smoothly without significant bouncing, and 3. the aircraft did not veer significantly to the left or right before it stopped rolling on the runway. After the data was collected, the video recordings of the landings were reviewed and any landings that did not meet the criteria for acceptable landings were discarded.

4. Autonomous flight

To test the viability of implementing the EXFA System into an autonomous flight scenario, the EXFA tractor aircraft was utilized (see Fig. 9). Using Mission Planner ground station software, the flight path was pre-programmed and stored on the computer. Two flight paths were created, one traveling counter-clockwise and one traveling clockwise, with respect to the runway, to allow the aircraft to takeoff into the wind regardless of the wind direction. The flight path was then uploaded to the Pixhawk 2.1 flight controller on the aircraft.

The clockwise flight path is shown in Fig. 13a. Both flight paths follow the same structure. After the aircraft was armed and the autonomous flight had been initiated, the aircraft automatically accelerated the motor to a high RPM setting to create maximum thrust output and maintained a straight heading until the final takeoff altitude of 15 m of was reached. The EXFA control surfaces were positioned in the flap position with 40° control surface deflection before takeoff and maintained that position for the majority of the flight. Subsequently, the aircraft maintained a constant altitude of 15 m and followed a curved flight path around the field to set up the aircraft for the landing approach. Once on the landing approach, the aircraft maintained a linear descent rate to direct the aircraft to land at a specified point on the runway. Flights were conducted with EXFA System in the neutral position, flap position, and flap position with spoiler position deployed 3 m from the touchdown point. During testing, the flight path and timing of waypoints and control surface deployments was adjusted until the aircraft could repeatedly complete a successful fully autonomous flight.

To compare the effects of the EXFA System, autonomous flights were conducted until two successful autonomous landings were recorded for each EXFA position on the final approach. Video recordings of the flights were reviewed, and the best landing with the most stable and consistent landing approach was selected for each EXFA position. The tree line in the camera view was selected as landing altitude reference, once the aircraft crossed this reference altitude, the vertical position on the runway was recorded. Then, the landing position on the runway is recorded, to calculate horizontal distance traveled during the landing approach. This procedure is shown in Fig. 13b. Additionally, airspeed and altitude data were recorded by the flight recorder at approximately 10Hz. The approach distance (*d*) was approximated by summing up the distance traveled between the data acquisition times using the airspeed (V_{plane}) and time (*t*) between individual data samples (*i*) according to Eq. 10:

$$d = \sum_i \bar{V}_{plane,i} \cdot \Delta t_i = \sum_i \frac{V_{plane,i} + V_{plane,(i-1)}}{2} \cdot \left(t_i - t_{(i-1)}\right) \qquad (10)$$

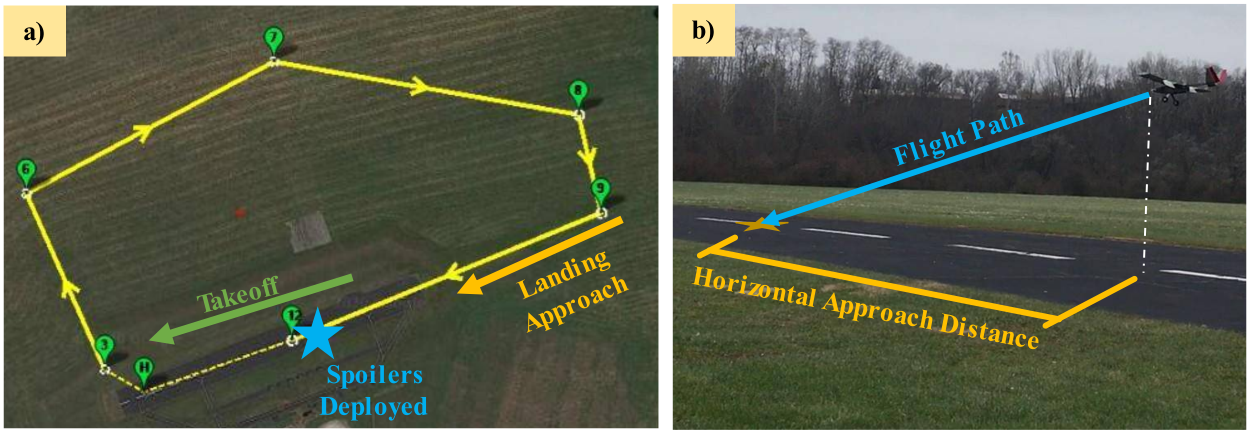

Fig. 13 a) Satellite view of flight area showing the clockwise flight path and the position of in-air spoiler position deployment, and b) diagram of landing approach.

IV. Results

A. Wind Tunnel Experimental Results

The baseline wing was initially compared to the hinged flap, slotted Fowler flap, EXFA, HEXFA, and HEXFA-VG wings at 0° control surface deflection for all angles of attack in the testing matrix (see Table 1). The average difference for the lift coefficient, C_L between the baseline and difference flap system was around 5%. Thus, the C_L for the 0° control surface deflection position on all wings was considered to be the same as the baseline. Due to the function of flap systems as lift increasing devices, only the C_L of the flap positions were compared. The datapoints displayed represent the mean value of each test case, with standard deviations shown as error bars.

1. Flap Configuration

The lift coefficient data for each flap position is shown in Fig. 14a and Fig. 14b. For the 40° flap position (Fig. 14b), the HEXFA System exhibited around 6% higher C_L values at low angles of attack (-2° to 8°) compared to the other flap mechanisms. The slotted Fowler flap was the only wing that did not exhibit a severe stall. Except for the EXFA wing, the other systems stalled very similarly under 16° angle of attack. Significant difference is found at the 20° flap position (Fig. 11a). The EXFA system has a very similar performance as the conventional flap system, while the HEXFA and HEXFA-VG wings exhibited approximately 25% greater C_L values at low angles of attack (-2° to 6°). This suggests that the HEXFA and HEXFA-VG Systems provide greater lift increase compared to currently used flap systems at medium deflection angles.

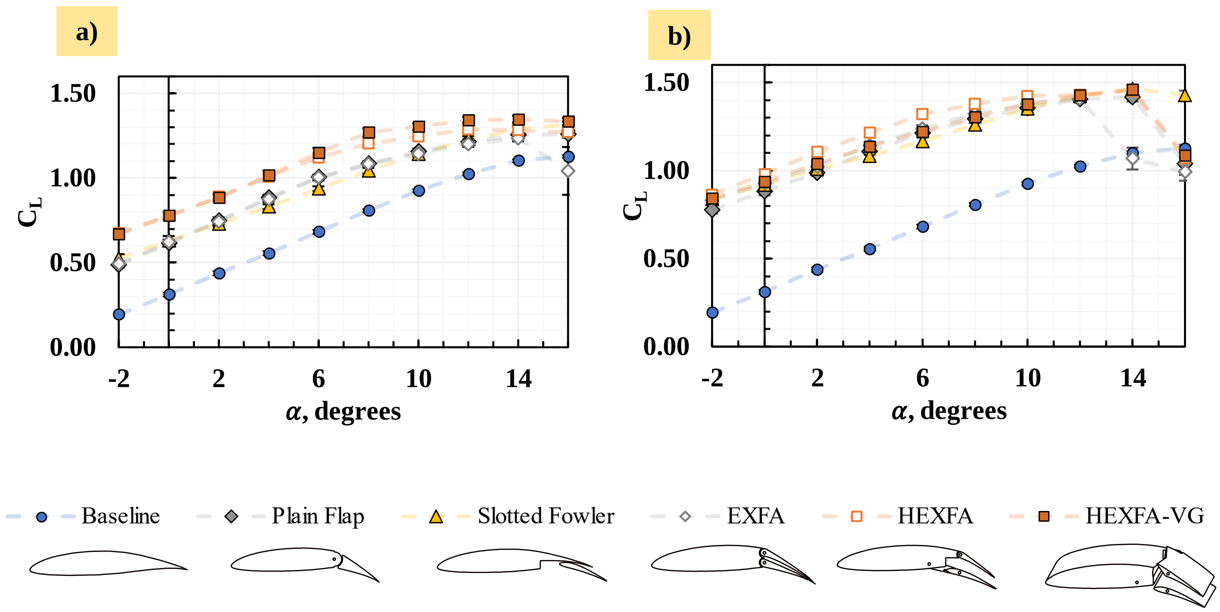

Fig. 14 Force data for a) 20° and b) 40° flap positions.

Additionally, the CFD data was analyzed. The C_L for the baseline airfoil and the slotted Fowler flap and HEXFA System airfoils at 20° flap deflection angle is shown in Fig. 15, compared with the corresponding wind tunnel data. The baseline C_L from the CFD simulations was very similar to the wind tunnel data for low angles of attack, which suggested that the CFD simulations matched the wind tunnel testing. The slotted Fowler flap C_L from the CFD simulation was slightly higher than the wind tunnel data. This was most likely caused by slight deformation in the flap control surface on the slotted Fowler flap wing during wind tunnel testing, which could have reduced its lift generation. Between the CFD simulations, the HEXFA System C_L is almost identical to the slotted Fowler flap C_L for angles of attack under 6° and performs slightly better at high angles of attack. Therefore, it was concluded that the combined wind tunnel and CFD data suggests that the HEXFA System has similar lift performance to the slotted Fowler flap at 20° flap deflection. Given the high-lift characteristics of the slotted Fowler flap [6], the similarity in flap performance between the slotted Fowler flap and the HEXFA System might indicate that this system could be an alternative to the slotted Fowler flap with a simpler mechanism while still retaining similar lift performance.

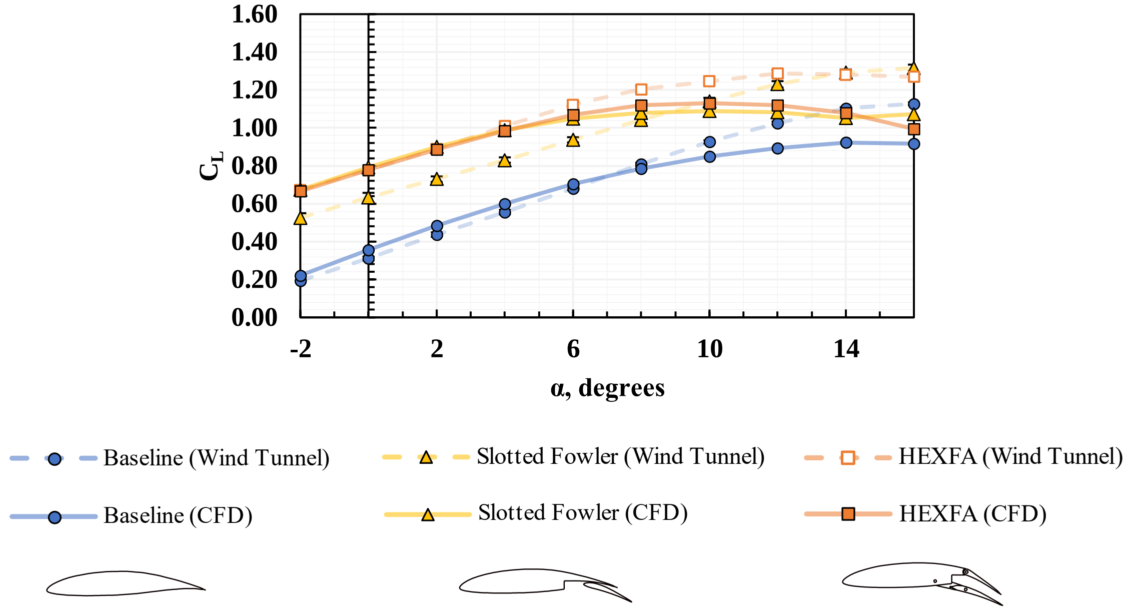

Fig. 15 CFD CL data for the baseline, slotted Fowler flap, and HEXFA System airfoils compared with the wind tunnel data at 20° flap deflection angle.

The normalized velocity contours for the wings were also analyzed for the slotted Fowler flap and the HEXFA System airfoils at 6° angle of attack (Fig. 16). These contours revealed an interesting finding regarding the slot placement of the two flap systems. For the slotted Fowler flap, there was a low velocity and recirculation region in front of the flap slot. Conversely, the flap slot on the HEXFA System was projected into the attached flow on the lower surface of the wing. The velocity contours suggest that the HEXFA System's placement of the flap slot outside of the wing profile could provide better slotted flap performance than the slotted Fowler flap.

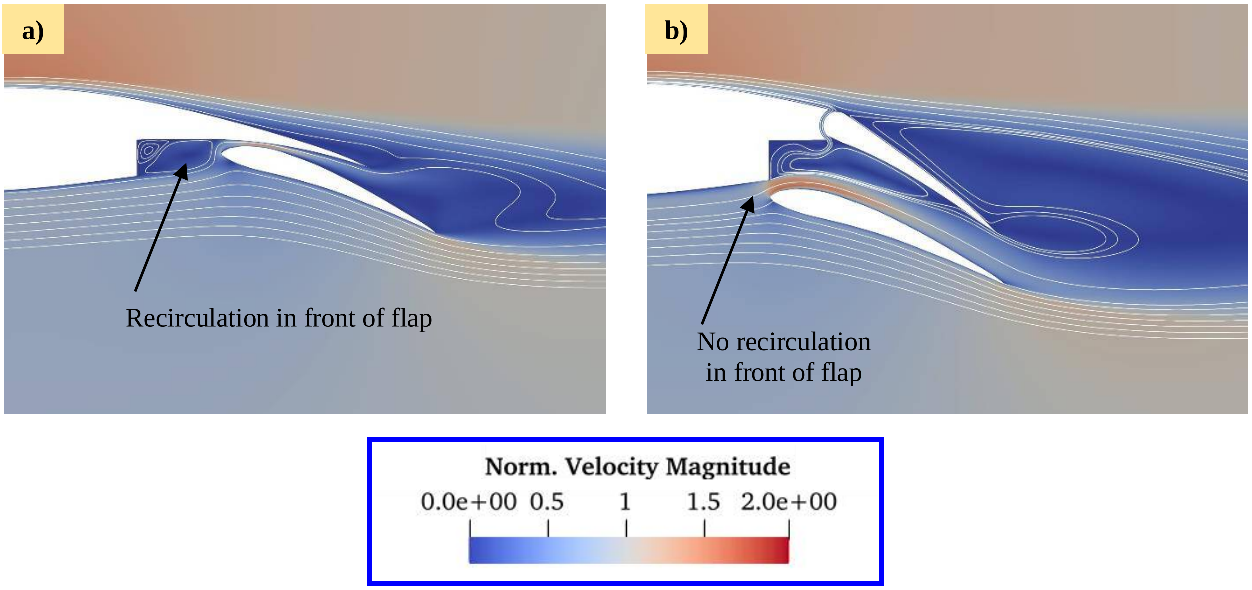

Fig. 16 Velocity contours for a) slotted Fowler flap and b) HEXFA System airfoils at 6° angle of attack, normalized by the freestream velocity.

2. *Aileron configuration*

The change in rolling moment ($\Delta\tau_x$) for different aileron position is shown in Fig. 17a and Fig. 17b. For both deflection angles, the HEXFA-VG wing exhibited the lowest performance, retaining at most 50% of the change in rolling moment of the plain flap wing, and the EXFA wing exhibited the best performance, retaining over 80% of the change in rolling moment of the plain flap wing for most of the dataset. HEXFA wing retained around 65% of the aileron roll authority of the plain flap wing.

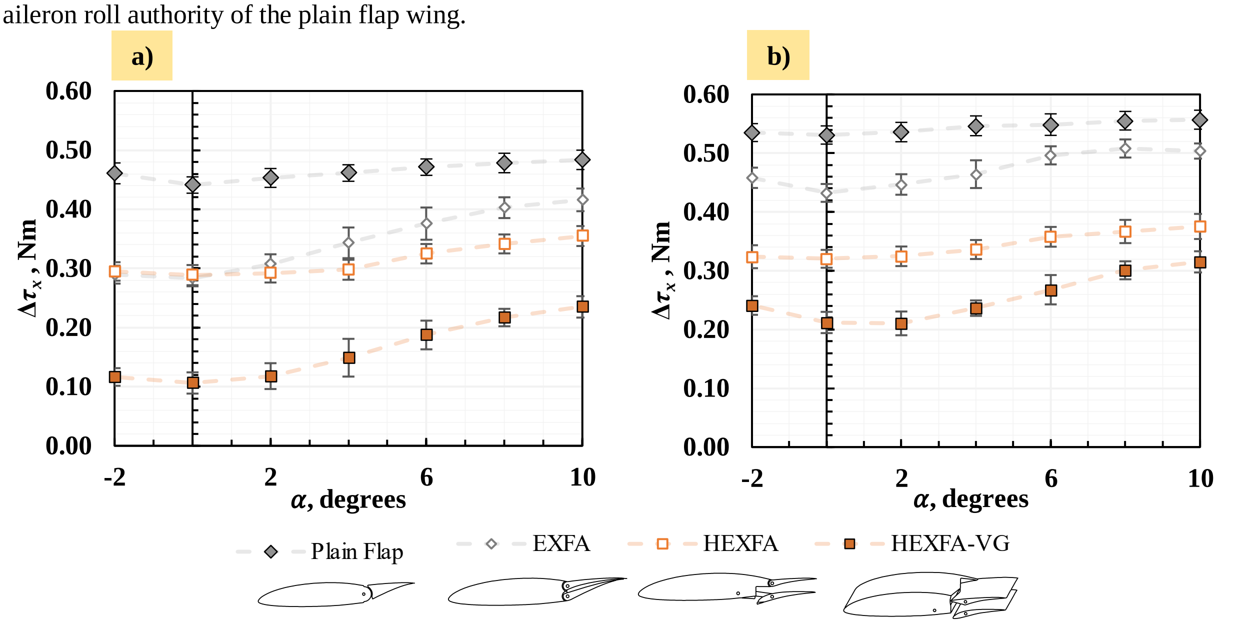

Fig. 17 Force data for a) 10° and b) 20° Aileron Position.

3. *Spoiler Configuration*

The change in C_D (ΔC_D) for the spoiler position is shown in Fig. 18a and Fig. 18b. Surprisingly, the HEXFA System exhibited the greatest increase in C_D for both deflection angles. Although the EXFA System provided a similar increase in C_D at the 40° spoiler deflection angle, the change in drag coefficient was lower than the HEXFA System for the 20° spoiler deflection angle. Based on the performance in lift, rolling moment and drag generation, the HEXFA System performs the best out of all the proposed designs as an all-around flap, aileron, and spoiler control system.

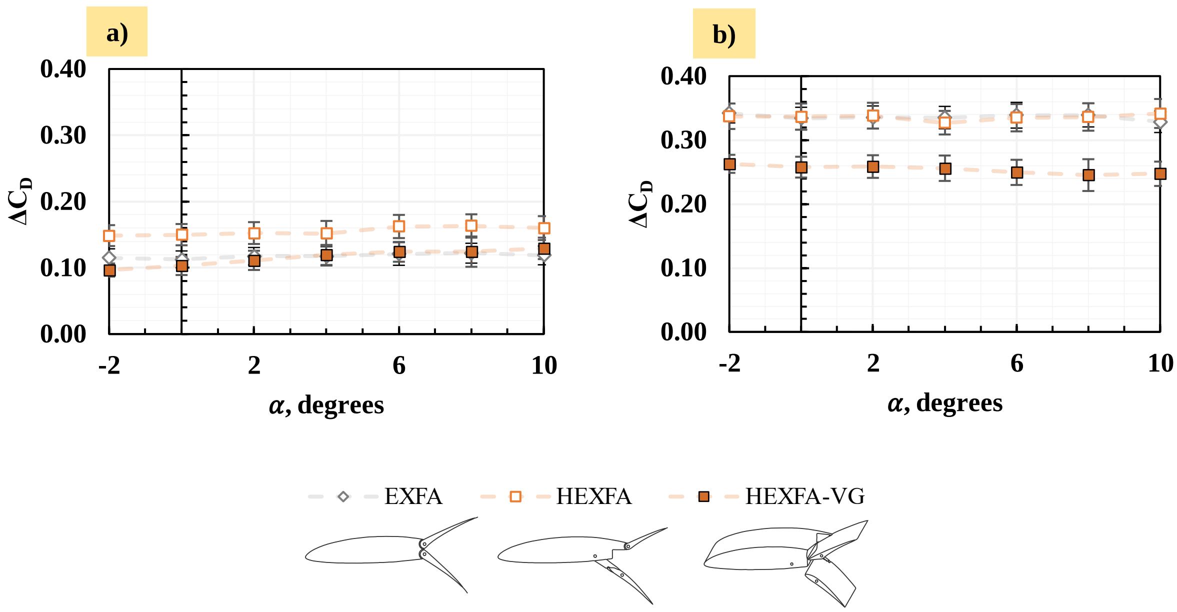

Fig. 18 Force data for a) 20° and b) 40° Spoiler Position

B. UAV Flight Test Results

1. Drag Data

To investigate the effect of the EXFA System on the in-flight characteristics of a small UAV, the drag data was analyzed. During the flight passes, which were all conducted on the same day, the weather conditions were optimal. From ground-based wind speed measurements and visual observations, there was no detectable wind throughout the testing and the overall conditions (cloud cover, etc.) stayed the same for the entirety of the data collection, which prevented the need to adjust for any wind interference in the data. The change in C_D (ΔC_D) is shown in Fig. 19, plotted against the change in C_D from the wind tunnel data for the flap and spoiler positions, all with 40° control surface deflections. At 0° angle of attack, the change in C_D from the flight test data was around 40% higher than the wind tunnel data for both the flap and spoiler positions. However, the spoiler position change in C_D was double that of the flap position for both the flight test and wind tunnel data. The larger magnitude of the flight test results was most likely due to the larger EXFA chord length on the flight test aircraft (0.45c) than on the wind tunnel wing (0.33c). The proportional similarity between the datasets suggests that the EXFA System performed as expected in the flight tests and verified its functionality in a full aircraft system.

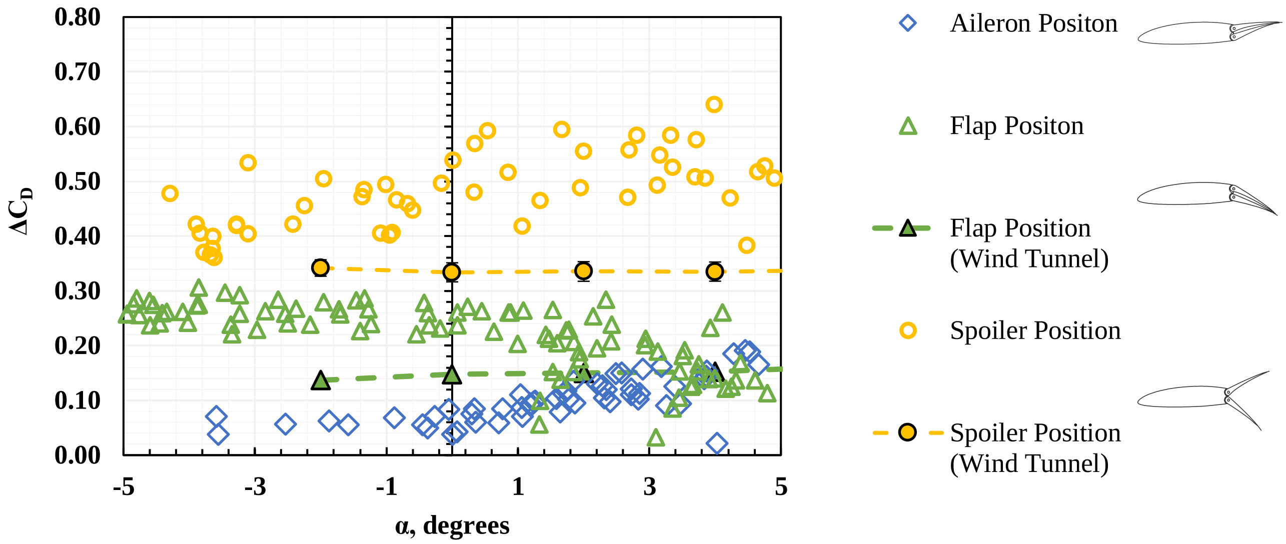

Fig. 19 Change in C_D from flight test results plotted against wind tunnel results for 40° control surface deflections.

2. Landing Distance Data

For the UAV flight tests, four landings were conducted without deploying the EXFA system, five landings were conducted with only the EXFA flaps deployed, and eleven landings were conducted utilizing the whole EXFA system. The results from all flights are shown in Fig. 20. Despite the low number of landings conducted, there was a statistically significant difference (ρ<0.05) between all of the datasets. The full EXFA system decreased the ground roll distance of the aircraft by over 50% compared to the baseline landing distance without utilizing the EXFA System (Fig. 20). The EXFA flaps alone only reduce the landing distance by 25% when compared to the baseline (Fig. 20). This shows that the EXFA System could be significantly more effective than just flaps at shortening an aircraft's landing distance.

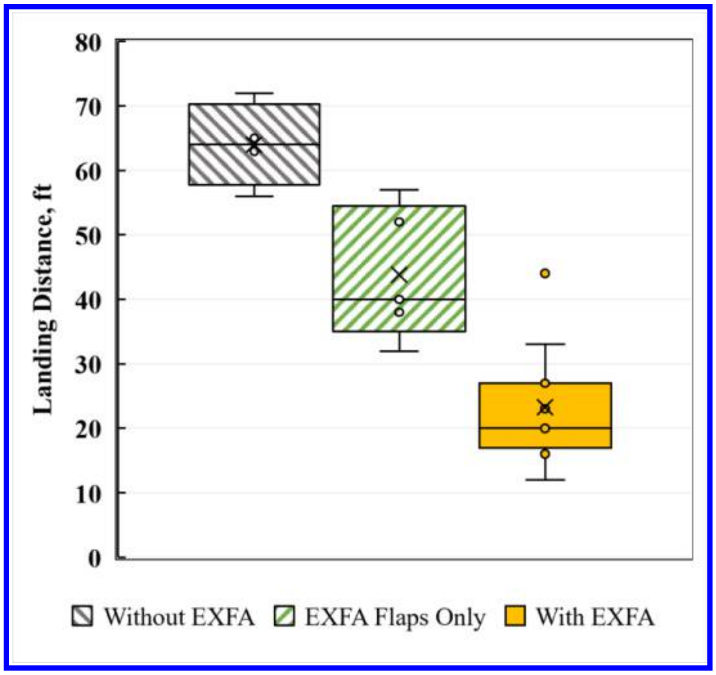

Fig. 20 box and whisker plot of landing distances.

3. Autonomous Flight

During testing, the aircraft was successfully able to fully autonomously takeoff with the EXFA flap position deployed, fly around the field, deploy the EXFA spoiler position moments before landing, and land and stop on the runway without any pilot intervention (Fig. 21a). This suggests that the EXFA System could be implemented in other autonomous UAVs and be utilized in more complex autonomous flight scenarios.

Throughout the testing, two successful autonomous landings were recorded for each EXFA position on the final approach (neutral position, EXFA flap position, and EXFA spoiler position), the best landing for each EXFA position was selected, and the horizontal final approach distance was calculated. The horizontal approach distance with the EXFA system in the neutral position was around 22.5 m long, and the distance with the EXFA flap position was 18 m long. However, the horizontal approach distance for the EXFA spoiler position was around 7.5 m, which was under half of both the neutral and flap position approach distances. A qualitative representation of the landing approach paths is shown in Fig. 21b with the horizontal approach distance labeled. The calculated approach distance from the flight recorder data is shown in Fig. 22. To match the visual landing distance approximations, an initial altitude of 1.75 m was used as the maximum approach height. The calculated approach distance for the neutral position was between 10 to 18 m, and the approach distance for the EXFA flap position was between 9 to 13 m. These values were slightly lower than the visually estimated distances. However, the calculated approach distance for the EXFA Spoiler position was between 2 to 9 m, which was similar to the visual estimate and still around half of the length of the neutral position approach distance. This supports the previous findings that the EXFA System could shorten landing approach lengths and could allow aircraft to utilize steeper descent slopes in a controlled manner.

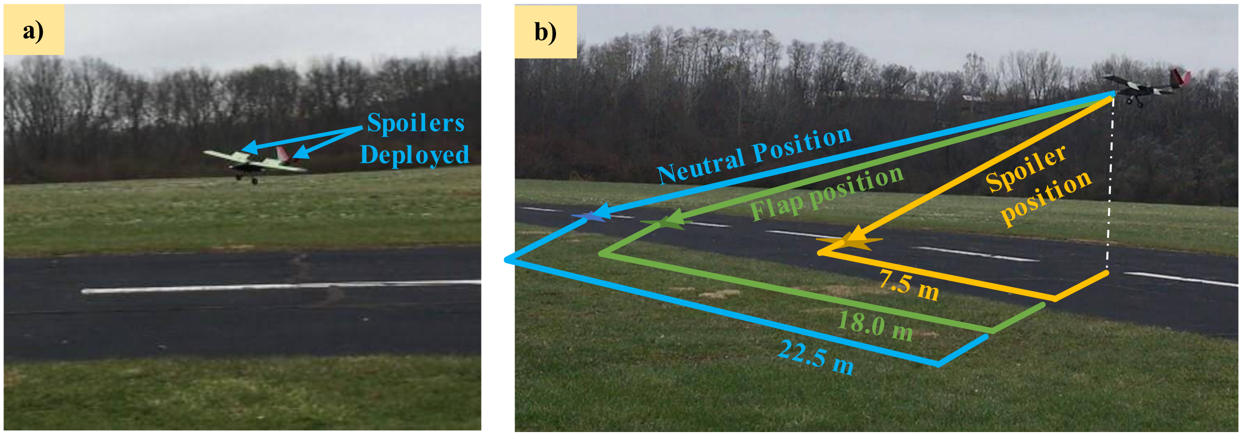

Fig. 21 a) Picture of aircraft with spoilers deployed moments before touching down on the runway and b) qualitative illustration of difference in approach distance.

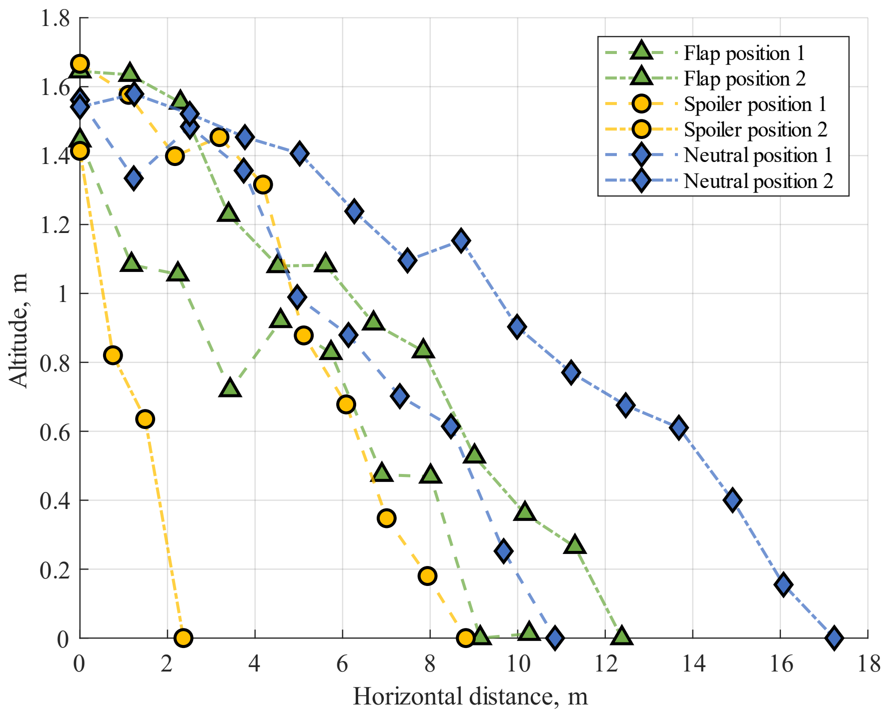

Fig. 22 Graph of altitude as a function of calculated approach distance from an initial height of 1.75m for landings 1 and 2 for each EXFA position.

V. Conclusion

From the wind tunnel testing and CFD simulations, the results strongly suggest that all of the three proposed control surface designs (EXFA, HEXFA, and HEXFA-VG Systems) can provide flap, aileron, and spoiler functionalities. Between these systems, the results suggest that the HEXFA System provides the best all-around performance due to its high-lift characteristics in the flap position, the mid-range roll authority in the aileron position, and high drag capabilities in the spoiler position. The results also suggest that the HEXFA System exhibits comparable lift characteristics and better slot placement to a slotted Fowler flap at low angles of attack. From the flight data for both high-endurance small UAVs equipped with the EXFA system, the results matched the predictions from the wind tunnel and CFD findings and suggested that the EXFA System can be implemented successfully in a full aircraft system and could be more effective than plain flaps at shortening the landing approach and reducing the ground roll distance. Given similarity between EXFA and HEXFA, the results suggest that HEXFA could also be implemented on aircraft with similar success. This could benefit high endurance aircraft by providing shorter takeoff and landing capabilities, three functions in one control surface mechanism, and deployment in autonomous flight, enabling these aircraft to be positioned in more locations, perform in more extreme conditions, and provide greater benefits to all types of operations.

VI. Future Work

Future work needs to be conducted to further investigate the aerodynamic properties of the designed systems. Additionally, further quantitative analysis of the autonomous test flights could be performed. Future work also needs to be directed towards implementing the HEXFA System into a high-endurance UAV to test its viability and integrability into the aircraft's control system. Additionally, research and testing could be performed to investigate efficacy of these system designs in non-conventional aircraft designs, including flying wings, delta wings, and blended-body aircraft.

Acknowledgments

The authors would like to thank Dr. Aaron Altman for his guidance and advice on project and Frank Beafore for providing advice and the UAV airframe for the drag data and autonomous flight testing.

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